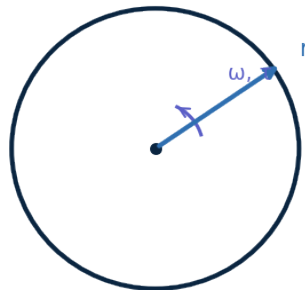


Rotational Motion

JEE Physics — Notes, Formula Sheet, Practice Problems & Diagram

1. Notes

- Angular displacement (θ), angular velocity ($\omega = d\theta/dt$), and angular acceleration ($\alpha = d\omega/dt$) describe rotational motion, analogous to x , v , a in linear motion.
- Torque $\tau = r \times F$ (cross product); magnitude $= rF \sin(\theta)$. Torque produces angular acceleration: $\tau = I\alpha$.
- Moment of inertia I depends on mass distribution about the axis. Common values: solid sphere $I = (2/5)MR^2$, solid cylinder/disc $I = (1/2)MR^2$, ring $I = MR^2$, rod about center $I = (1/12)ML^2$.
- Parallel axis theorem: $I = I(\text{cm}) + Md^2$. Perpendicular axis theorem (for planar bodies): $I(z) = I(x) + I(y)$.
- Angular momentum $L = I\omega$. In the absence of external torque, angular momentum is conserved (law of conservation of angular momentum).
- Rotational kinetic energy $= (1/2) I \omega^2$. For rolling without slipping, total KE $= (1/2)Mv^2 + (1/2)I\omega^2$, with $v = R\omega$.
- Condition for rolling without slipping: $v(\text{center}) = R\omega$, where R is the radius of the rolling body.



Particle in circular motion: radius vector r , angular velocity ω , and torque τ .

2. Formula Sheet

Torque	$\tau = r F \sin(\theta) = I \alpha$
Angular momentum	$L = I \omega$
Moment of inertia (disc)	$I = \frac{1}{2} M R^2$
Moment of inertia (ring)	$I = M R^2$
Moment of inertia (solid sphere)	$I = \frac{2}{5} M R^2$
Parallel axis theorem	$I = I_{cm} + M d^2$
Rotational KE	$KE = \frac{1}{2} I \omega^2$
Rolling KE (translation + rotation)	$KE = \frac{1}{2} M v^2 + \frac{1}{2} I \omega^2$

3. Practice Problems (22 questions)

1. A wheel starts from rest and attains an angular velocity of 20 rad/s in 5 s. Find its angular acceleration.
2. A disc of mass 2 kg and radius 0.5 m rotates about its central axis. Find its moment of inertia.
3. A force of 10 N is applied tangentially at the rim of a wheel of radius 0.2 m. Find the torque produced.
4. A torque of 20 N m acts on a body of moment of inertia 4 kg m². Find its angular acceleration.
5. A ring of mass 1 kg and radius 0.5 m rolls without slipping with a linear speed of 2 m/s. Find its total kinetic energy.
6. A solid sphere of mass 5 kg and radius 0.2 m rotates at 10 rad/s about its diameter. Find its rotational kinetic energy.
7. State and explain the law of conservation of angular momentum with one real-life example.
8. A rod of mass 3 kg and length 2 m rotates about an axis through its center perpendicular to its length. Find its moment of inertia.
9. Using the parallel axis theorem, find the moment of inertia of a disc of mass M and radius R about an axis through its edge, perpendicular to its plane.
10. A flywheel of moment of inertia 10 kg m² is rotating at 100 rad/s. Find its angular momentum.
11. A figure skater spinning with arms outstretched has moment of inertia 4 kg m² and angular velocity 2 rad/s. She pulls her arms in, reducing her moment of inertia to 2 kg m². Find her new angular velocity.
12. A wheel of radius 0.5 m rolls without slipping with angular velocity 4 rad/s. Find the linear speed of its center.
13. Define the moment of inertia and explain why it depends on the axis of rotation, not just the mass of the body.
14. A torque of 50 N m acting on a wheel for 4 s changes its angular momentum from 0 to L. Find L.
15. A solid cylinder of mass 4 kg and radius 0.3 m rolls down an incline without slipping. Find its moment of inertia about its central axis.

- 16.** Two discs of moment of inertia 2 kg m^2 and 4 kg m^2 rotating at 10 rad/s and 5 rad/s respectively are brought into contact so they rotate together. Find their common angular velocity (assume they stick and combine, conserving angular momentum).
- 17.** A particle of mass 2 kg is moving in a circle of radius 3 m with speed 6 m/s . Find its angular momentum about the center.
- 18.** Explain the difference between center of mass and center of gravity.
- 19.** A uniform rod is pivoted at one end and released from a horizontal position. State whether its angular acceleration is maximum or minimum at the horizontal position, with reason.
- 20.** A ceiling fan takes 4 s to reach its full angular speed of 8 rad/s from rest. Find the number of revolutions it makes during this time (assume uniform angular acceleration).
- 21.** For a solid sphere rolling without slipping, show that its rotational KE is $(2/7)$ of its total kinetic energy.
- 22.** A wheel of moment of inertia 2 kg m^2 is rotating at 50 rad/s . A braking torque of 10 N m is applied. Find the time taken for the wheel to stop.

Answer Key

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| 1. $\alpha = 4 \text{ rad/s}^2$ | 2. $I = 0.25 \text{ kg m}^2$ |
| 3. $\tau = 2 \text{ N m}$ | 4. $\alpha = 5 \text{ rad/s}^2$ |
| 5. $\text{KE} = 4 \text{ J}$ | 6. $\text{KE} = 20 \text{ J}$ |
| 7. Conceptual - L is conserved when net external torque = 0; e.g., ice skater pulling arms in | 8. $I = 1 \text{ kg m}^2$ |
| 9. $I = (3/2) M R^2$ | 10. $L = 1000 \text{ kg m}^2/\text{s}$ |
| 11. $\omega = 4 \text{ rad/s}$ | 12. $v = 2 \text{ m/s}$ |
| 13. Conceptual - I depends on mass distribution relative to the chosen axis | 14. $L = 200 \text{ kg m}^2/\text{s}$ |
| 15. $I = 0.18 \text{ kg m}^2$ | 16. $\omega = 6.67 \text{ rad/s}$ (approx) |
| 17. $L = 36 \text{ kg m}^2/\text{s}$ | 18. Conceptual - same location unless in a strong non-uniform gravitational field |
| 19. Conceptual - angular acceleration is maximum at horizontal position (max torque due to gravity) | 20. $16 \text{ rad} \rightarrow 2.55 \text{ revolutions}$ (approx) |
| 21. Conceptual derivation - see notes above | 22. $t = 10 \text{ s}$ |

Note: conceptual/derivation questions are marked accordingly in the answer key. Numerical answers are rounded for clarity — show full working in your own solutions.